



AQ3.5: Activity Questions 5 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

1 point

Choose the set of correct options.

☐

The set $\{(1, 2, 3), (4, 5, 6), (7, 8, 9)\}$ is linearly dependent.

☐

The set $\{(1, 2, 3), (4, 5, 6), (5, 7, 9)\}$ is linearly independent.

☐

The set $\{(1, 2, 3), (0, 5, 6), (0, 0, 9)\}$ is linearly independent.

☐

The set $\{(1, 2, 3), (0, 0, 6), (0, 0, 9)\}$ is linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{(1, 2, 3), (4, 5, 6), (7, 8, 9)\}$ is linearly dependent.

The set $\{(1, 2, 3), (0, 5, 6), (0, 0, 9)\}$ is linearly independent.

1 point

If $ad - bc = 0$ then which of the following options are true?

☐

The set $\{(a, b), (c, d)\}$ is linearly dependent in R^2 .

☐

The set $\{(a, c), (b, d)\}$ is linearly dependent in R^2 .

☐

The set $\{(1, 0, 0), (0, a, b), (0, c, d)\}$ is linearly dependent in R^3 .

☐

The set $\{(1, 0, 0), (0, a, c), (0, b, d)\}$ is linearly dependent in R^3 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{(a, b), (c, d)\}$ is linearly dependent in R^2 .

The set $\{(a, c), (b, d)\}$ is linearly dependent in R^2 .

The set $\{(1, 0, 0), (0, a, b), (0, c, d)\}$ is linearly dependent in R^3 .

The set $\{(1, 0, 0), (0, a, c), (0, b, d)\}$ is linearly dependent in R^3 .

For $c \in \mathbb{Z}$, consider the set of three vectors $S = \{(1, c, 0), (c, 0, c), (2c, 3c, 4c)\}$ in R^3 with usual addition and scalar multiplication. Find the value of c such that the given set is linearly dependent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

1 point



Assume the set $\{(a_1, b_1, c_1), (a_2, b_2, c_2), (a_3, b_3, c_3)\}$ $\{(a_1, b_1, c_1), (a_2, b_2, c_2), (a_3, b_3, c_3)\}$ is given to be linearly independent and consider the following system of linear equations

$$\begin{aligned} a_1 x + b_1 y + c_1 z &= 1 \\ a_2 x + b_2 y + c_2 z &= 2a_1 x + b_1 y + c_1 z = 1a_2 x + b_2 y + c_2 z = 2a_3 x + b_2 y + c_3 \\ a_3 x + b_2 y + c_3 z &= 3. \end{aligned}$$

$$z = 3.$$

If $AX = b$ is matrix representation of the above system of linear equations, where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, then which of the following option(s) is(are) true?

- ☐ The system of linear equations has a unique solution.
- ☐ The system of linear equations has no solution.
- ☐ The system of linear equations have infinitely many solutions.
- ☐ The coefficient matrix AA is invertible.
- ☐ The coefficient matrix AA is not invertible.
- ☐ Any one of the above options cannot be concluded using the given information.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The system of linear equations has a unique solution.

The coefficient matrix AA is invertible.

1 point

Assume the set $\{(a_1, b_1, c_1), (a_2, b_2, c_2), (a_3, b_3, c_3)\}$ $\{(a_1, b_1, c_1), (a_2, b_2, c_2), (a_3, b_3, c_3)\}$ is given to be linearly dependent and consider the following system of linear equations

$$\begin{aligned} a_1 x + b_1 y + c_1 z &= 1 \\ a_2 x + b_2 y + c_2 z &= 2a_1 x + b_1 y + c_1 z = 1a_2 x + b_2 y + c_2 z = 2a_3 x + b_2 y + c_3 \\ a_3 x + b_2 y + c_3 z &= 3. \end{aligned}$$

$$z = 3.$$

If $AX = b$ is matrix representation of the above system of linear equations, where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, then which of the following option(s) is(are) true?

- ☐ Either the system of linear equations has a unique solution or no solution.
- ☐ Either the system of linear equations has no solution or infinitely many solutions.
- ☐ Either the system of linear equations have infinitely many solutions or a unique solution.
- ☐ The system of linear equations has a unique solution.
- ☐ The coefficient matrix AA is invertible.
- ☐ The coefficient matrix AA is not invertible.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Either the system of linear equations has no solution or infinitely many solutions.

The coefficient matrix AA is not invertible.

If SS is a linearly independent set in vector space R^3 with respect to usual addition and scalar multiplication, then find the maximum possible cardinality of SS .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

Level 2

1 point

Let the set $\{(a, b), (c, d)\}$ be a linearly independent subset of R^2 . Choose the set of correct options.

☐

$\{(a, b, 0), (c, d, 0)\}$ must be a linearly independent subset of R^3 .

☐

$\{(a, b, 0), (c, d, 0), (0, 0, 1)\}$ must be a linearly independent subset of R^3 .

☐

$\{(a, b, 0), (c, d, 0), (1, 0, 0)\}$ must be a linearly independent subset of R^3 .

☐

$\{(a, b, 0), (c, d, 0), (0, 1, 0)\}$ must be a linearly independent subset of R^3 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(a, b, 0), (c, d, 0)\}$ must be a linearly independent subset of R^3 .

$\{(a, b, 0), (c, d, 0), (0, 0, 1)\}$ must be a linearly independent subset of R^3 .

1 point

Each column of an $n \times n$ matrix AA can be thought of as vectors in R^n . If a set SS consists of all the column vectors of AA , then which of the following is(are) true?

☐

SS is always linearly independent.

☐

SS is linearly independent if $\det(A) \neq 0$.

☐

SS is linearly independent if $\det(A) = 0$.

☐

If $\det(A) = 1$, then SS is linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

SS is linearly independent if $\det(A) \neq 0$.

If $\det(A) = 1$, then SS is linearly independent.

1 point

Suppose it is known that the three vectors $v_1 = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$, $v_2 = \begin{bmatrix} d \\ e \\ f \end{bmatrix}$, $v_3 = \begin{bmatrix} g \\ h \\ i \end{bmatrix}$ are linearly dependent in R^3 . Which of the following is correct?

☐

Some linear combination of a, b, c must be 0.

☐

Some linear combination of a, d, g must be 0.

☐

Some linear combination of d, e, f must be 0.

☐

Some linear combination of c, f, i must be 0.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Some linear combination of a, b, c must be 0.

Some linear combination of a, d, g must be 0.

Some linear combination of d, e, f must be 0.

Some linear combination of c, f, i must be 0.

If vectors $(2x, 4, -6), (-6, -3x, 2x), (-1, x-3, x+2)$ from the vector space R^3 with usual addition and scalar multiplication, are linearly dependent and $x \in [2, 5]$, then find the value of x .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point